TOPOLOGICAL RINGS HAVE LIMITS, FORMED ON UNDERLYING SPACES AND RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The reduction of limits to products and equalizers. Invoked at the start of the counted proof (source lines 6139-6141) as a cross-chapter pointer. Restricts the assertion to those two shapes of limits.

Source reference. Original reference categories-lemma-limits-products-equalizers: sources/stacks-topology/pinned-original-source.tex, source lines 2582–2582. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-limits-products-equalizers, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *The category of topological rings has limits and limits commute with the forgetful functors to (a) the category of topological spaces and (b) the category of rings.*

Proof. It is enough to prove the existence and commutation for products and equalizers, see Categories, Lemma categories-lemma-limits-products-equalizers. Let R_i, $i \in I$ be a collection of topological rings. Take the usual product $R = \prod R_i$ with the product topology. Since $R \times R = \prod (R_i \times R_i)$ as a topological space (because products commutes with products in any category), we see that addition and multiplication on R are continuous. Let $a, b : R \to R'$ be two homomorphisms of topological rings. Then as the equalizer we can simply take the equalizer of a and b as maps of topological spaces, which is the same thing as the equalizer as maps of rings endowed with the induced topology. $\square$

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